PAPER ID-311299

Subject Code: BOE312

Roll No: | | | | | | | | | | | | | |

**BTECH**
**(SEM III) THEORY EXAMINATION 2023-24**
**LASER SYSTEM AND APPLICATIONS**

**TIME: 3HRS** **M.MARKS: 70**

**Note: 1.** Attempt all Sections. If require any missing data; then choose suitably.

## SECTION A

**1. Attempt *all* questions in brief.** **2 x 7 = 14**

| | |
|---|---|
| a. | Define Spatial coherence. |
| b. | What do you men by Energy Level? |
| c. | Define Resonators. |
| d. | Define Loop gain. |
| e. | What is Semiconductor diode laser? |
| f. | Why pumping is necessary for laser action? |
| g. | List the safety consideration of Laser. |

## SECTION B

**2. Attempt any *three* of the following:** **7 x 3 = 21**

| | |
|---|---|
| a. | Differentiate between Longitudinal Coherence length, Transverse Coherence length |
| b. | What is optical cavity? Calculate the loop gain with and without losses for optical cavity. |
| c. | Differentiate between continuous wave laser and pulse laser. Discuss various pulsating technique for a laser. |
| d. | Give the advantage and disadvantage of laser in material processing. |
| e. | What are solid state lasers? Describe construction, working and application of alexandrite lasers. What is its advantage? |

## SECTION C

**3. Attempt any *one* part of the following:** **7 x 1 = 7**

| | |
|---|---|
| a. | Define Laser. Give the Properties, Advantages and Disadvantages of laser beams |
| b. | Define Spontaneous Emission and Stimulated emission processes with its characteristics. |

**4. Attempt any *one* part of the following:** **7 x 1 = 7**

| | |
|---|---|
| a. | Define Two, three and four level pumping schemes. |
| b. | What is Gain in lasers? Also define Gain factor. |

**5. Attempt any *one* part of the following:** **7 x 1 = 7**

| | |
|---|---|
| a. | What are the main components of laser? Briefly explain each component. |
| b. | What are the advantages of four level lasers over three level lasers? Derive an expression for threshold pumping power required to start laser action in three level laser systems. |



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**6. Attempt any *one* part of the following: 7 x 1 = 7**

| | |
|---|---|
| a. | Describe the construction and working of $CO_2$ laser. A gas laser is generating a laser beam of 4 mW power. Calculate the number of photons emitted by the laser. The wave length of the emitted radiation is 680 nm. |
| b. | With suitable schematic diagram explain the working of dye laser. How dye lasers are tuned? |

**7. Attempt any *one* part of the following: 7 x 1 = 7**

| | |
|---|---|
| a. | How lasers are useful in drilling and cutting. Discuss applications of lasers in melting. |
| b. | Discuss the principle of holography. Explain construction and reconstruction of hologram. |

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